

Polynomial Regression

ML Assignment 1 | Tanmay Dixit | BT2024016

Data and objective

Separate polynomial regression models predict the Net Power Score (var1) and Thermal Anomaly Score (var2). Each supplied training file has 1,000 labelled rows. Each test file has 1,000 rows without target labels. All six inputs are used for var1 and all three for var2. The four files contain no missing values. Original test-row order is preserved.

Polynomial models

Each term has a total degree equal to the sum of its exponents. For example, $x_1^2 x_2$ has total degree 3. The search includes powers and interactions, with degree capped at 10 for var1 and 20 for var2. Monomial and Legendre bases are compared. Both represent polynomials. A fixed average of polynomial fits is also a polynomial, with degree no greater than the largest component degree. No tree, kernel with non-polynomial terms, or neural network is used.

Model selection

A seed-42 split gives 800 development rows and 200 holdout rows per problem. Broad searches on the development rows compare ordinary least squares, Ridge, Lasso, relaxed Lasso, and automatic relevance determination (ARD) regression. ARD learns a separate shrinkage level for each coefficient. Expanded columns are standardized using only the fitting fold. Sparse term selection and any coefficient refit also stay inside that fold.

The all-feature expanded Ridge search covers degrees 1-6 for var1 and 1-12 for var2. Further sparse searches test var1 degrees 4, 5, 6, 8, 10 and var2 degrees 6, 8, 10, 12, 16, 20, followed by a finer penalty grid near the leading degrees. Ridge basis and degree-penalty searches extend to degree 8 and 20, respectively. The complete grids and attempted numerical fits are recorded with the code.

Nine individual finalists and two fixed equal-weight averages are compared per problem. Each is evaluated with three five-fold partitions of the same 800 development rows (seeds 42, 137, 2026). The lowest mean MSE over these 15 folds determines the final choice. These are model-selection scores, not an unbiased final test estimate. Repeated folds overlap.

Validation and final fitting

The chosen settings are fixed before the final holdout evaluation. The 200-row holdout was examined during preliminary experiments and excluded from subsequent tuning, so it is a reused validation set. After recording MSE and R-squared, the fixed models are refitted on all 1,000 labelled rows to create the submission predictions.

MSE is the mean squared prediction error; lower is better. R-squared is $1 - SSE/SST$; closer to 1 is better. Hidden-test scores cannot be computed because test targets were not supplied.

GitHub repository

<https://github.com/tdixit547/ml-assignment-1-bt2024016>

var1: Steam turbine optimization

Selected model

Selected degree: 5. The model averages two degree-5 monomial fits with equal weights: Lasso (alpha = 0.008) and ARD (maximum 500 iterations, tolerance 0.0001). Both use all six features and start from 461 polynomial terms, excluding the intercept. Coefficient shrinkage limits the influence of unsupported powers and interactions.

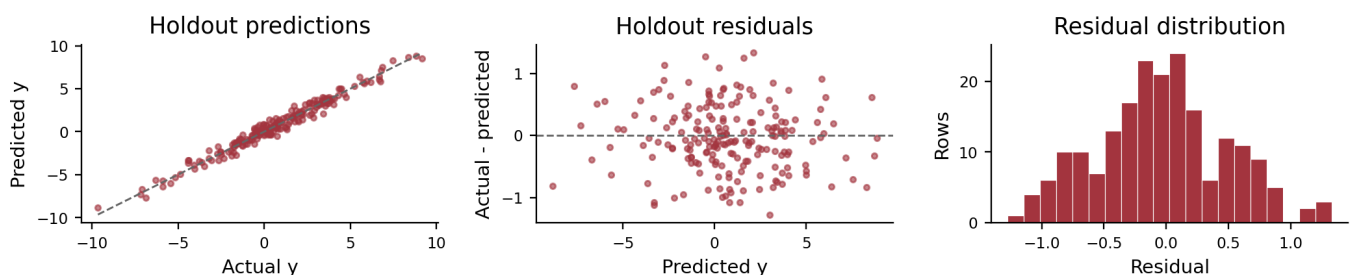
Measured results

Repeated development CV MSE: 0.299307. Individual fold MSE standard deviation: 0.027033. This dispersion is descriptive; it is not a confidence interval.

Selected-model holdout MSE: 0.278873. Holdout R-squared: 0.971026. The following table compares all 11 finalists on the same 15 development folds.

Model	Degree	Basis	Parameters	CV MSE	Fold SD
Lasso	5	monomial	alpha=0.005	0.306978	0.028680
Lasso	5	monomial	alpha=0.008	0.301518	0.026497
Lasso	5	monomial	alpha=0.01	0.303917	0.026279
Relaxed Lasso	5	monomial	alpha=0.015; relax=0.5; refit alpha=0.01	0.302024	0.024095
Relaxed Lasso	5	monomial	alpha=0.02; relax=0.5; refit alpha=0.01	0.307289	0.025154
Selected-term refit	5	monomial	alpha=0.03; relax=1; refit alpha=0.01	0.320594	0.035689
Lasso	6	monomial	alpha=0.01	0.324573	0.029279
ARD	5	monomial	max_iter=500; tol=0.0001	0.315411	0.029954
Ridge	5	legendre	alpha=0.1; degree penalty=2	0.457559	0.054652
Equal average (selected)	5	monomial	Lasso d5 alpha=0.008 + ARD d5; equal weights, component settings above	0.299307	0.027033
Equal average	5	monomial	Lasso d5 alpha=0.008 + relaxed Lasso d5 alpha=0.015; equal weights, component settings above	0.300787	0.025208

All 11 finalists use the same 15 development folds. Fold SD is descriptive dispersion. Degree is maximum total degree. The degree-penalty value p scales terms of degree k by k^{-p} . Relax is the weight on the selected-term Ridge refit; all expansions use fold-fitted standardisation.



Selection rationale and final predictions

Degree-5 Lasso had lower repeated CV MSE than the tested degree-6 Lasso and dense degree-5 Ridge finalists. Averaging the degree-5 Lasso and ARD fits gave the lowest mean CV MSE among the finalists. Its advantage over Lasso alone is small and does not establish a statistically significant difference. Test inputs are more often near the feature boundaries than training inputs, so random-split performance may not fully represent hidden-test performance.

Final training uses all 1,000 labelled rows. BT2024016_pred_var1.csv contains 1,000 finite predictions under the single header y. The saved JSON model supports inference without retraining.

var2: Thermal reservoir mapping

Selected model

Selected degree: 12. The model equally averages a degree-12 Legendre Ridge fit ($\alpha = 0.01$) and a degree-9 monomial relaxed-Lasso fit ($\alpha = 0.001$). For Ridge, a standardized term of total degree k is divided by k^2 before fitting, so high-degree standardized coefficients receive a stronger penalty. The relaxed-Lasso fit averages its Lasso coefficients with a Ridge refit ($\alpha = 0.01$) on selected terms. The two expansions contain 454 and 219 terms, excluding intercepts.

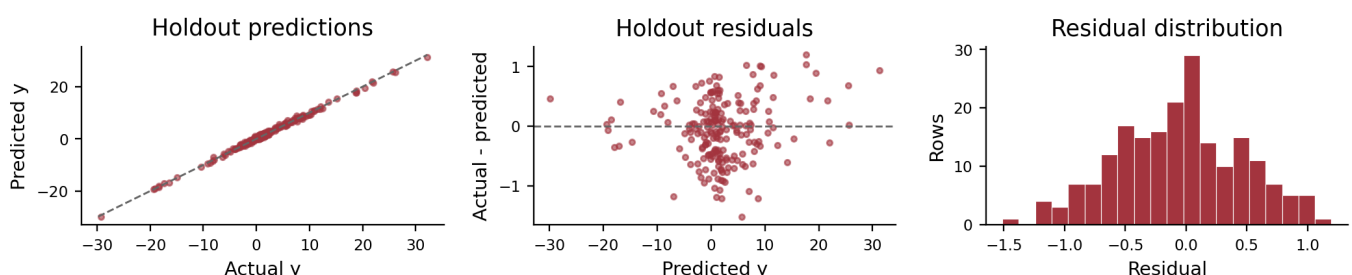
Measured results

Repeated development CV MSE: 0.278119. Individual fold MSE standard deviation: 0.040501. This dispersion is descriptive; it is not a confidence interval.

Selected-model holdout MSE: 0.274358. Holdout R-squared: 0.995233. The following table compares all 11 finalists on the same 15 development folds.

Model	Degree	Basis	Parameters	CV MSE	Fold SD
Ridge	12	legendre	$\alpha=0.01$; degree penalty=2	0.308728	0.054346
Ridge	14	legendre	$\alpha=0.01$; degree penalty=2	0.310564	0.053352
Ridge	11	legendre	$\alpha=0.01$; degree penalty=2	0.306841	0.058862
Ridge	8	legendre	$\alpha=0.316228$; degree penalty=1	0.312118	0.062212
Relaxed Lasso	9	monomial	$\alpha=0.001$; relax=0.5; refit $\alpha=0.01$	0.291013	0.045832
Selected-term refit	12	monomial	$\alpha=0.003$; relax=1; refit $\alpha=0.01$	0.317299	0.065098
Selected-term refit	8	legendre	$\alpha=0.02$; relax=1; refit $\alpha=0.01$	0.322435	0.042428
Selected-term refit	9	legendre	$\alpha=0.015$; relax=1; refit $\alpha=0.01$	0.316152	0.042961
Lasso	9	monomial	$\alpha=0.0005$	0.294363	0.053624
Equal average (selected)	12	legendre / monomial	Ridge d12 $\alpha=0.01$ + relaxed Lasso d9 $\alpha=0.001$; equal weights, component settings above	0.278119	0.040501
Equal average	12	legendre / monomial	Ridge d12 $\alpha=0.01$ + relaxed Lasso d9 $\alpha=0.001$ + selected-term refit d8 $\alpha=0.02$; equal weights, component settings above	0.278359	0.035818

All 11 finalists use the same 15 development folds. Fold SD is descriptive dispersion. Degree is maximum total degree. The degree-penalty value p scales terms of degree k by k^{-p} . Relax is the weight on the selected-term Ridge refit; all expansions use fold-fitted standardisation.



Selection rationale and final predictions

Degree-9 relaxed Lasso and degree-12 Ridge were competitive individual candidates. Their fixed average had lower mean repeated CV MSE than either component and was the best of the tested finalists. The resulting polynomial has maximum total degree 12. Hidden-test performance remains unknown.

Final training uses all 1,000 labelled rows. BT2024016_pred_var2.csv contains 1,000 finite predictions under the single header y. The saved JSON model supports inference without retraining.